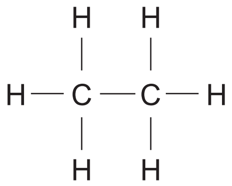
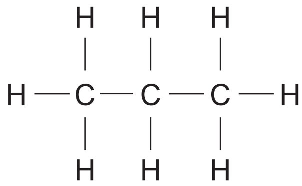
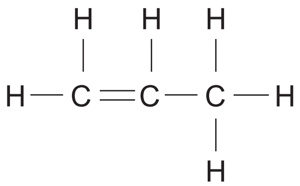
2. *(a)* The structural formulae of four carbon compounds are shown below.



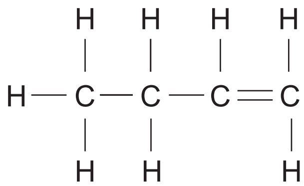
A



B



C

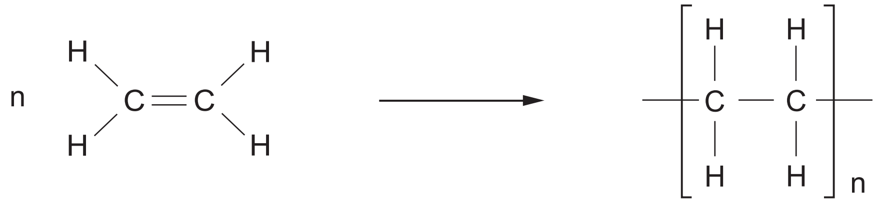


D

Complete the table by choosing the **letter A**, **B**, **C** or **D** which represents the structural formula of the named compounds. [2]

Name	Molecular formula	Structural formula
ethane	C_2H_6	....
propene	C_3H_6	....

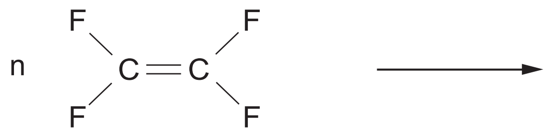
(b) The equation below shows the formation of polythene from ethene.



ethene polythene

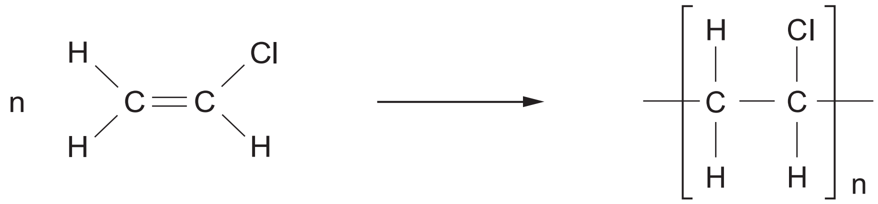
Use this information to complete parts (i) and (ii).

(i) Complete the equation for the formation of polytetrafluoroethene (PTFE) from tetrafluoroethene. [1]



tetrafluoroethene polytetrafluoroethene (PTFE)

(ii) Name monomer **E**. [1]

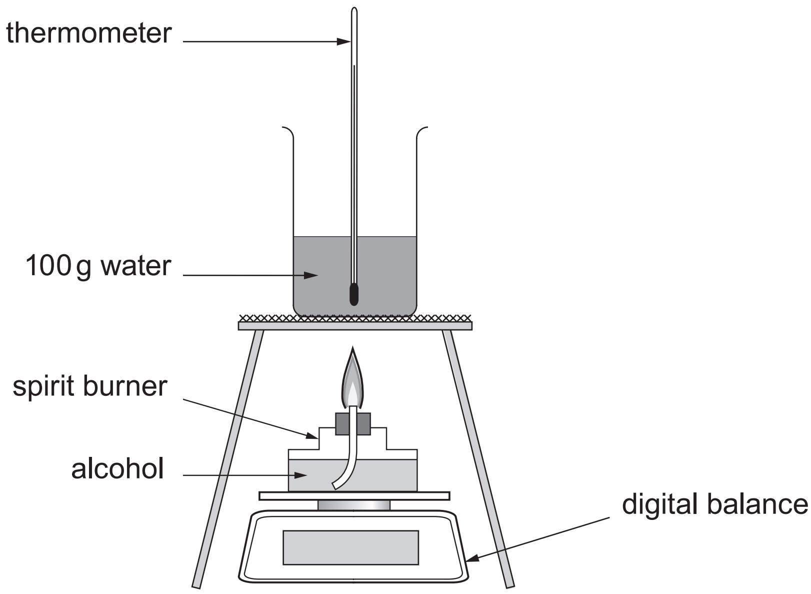


E polyvinylchloride (PVC)

E ....

Examiner only

(c) A student carried out an investigation to find which alcohol gives out the most energy when burned.



100 g of water was heated by burning 1 g of each alcohol.

Alcohol	methanol	ethanol	propanol
Formula	CH_3OH	C_2H_5OH	C_3H_7OH
Relative molecular mass (M_r)	?	46	60
Temperature of water before heating (°C)	20	21	20
Temperature of water after heating (°C)	45	58	60
Energy given out (J)	?	15 700	17 200

(i) Calculate the relative molecular mass (M_r) of methanol, CH_3OH. [2]

A_r(H) = 1 A_r(C) = 12 A_r(O) = 16

M_r = ...

 (3410U20-1)

Examiner only

(ii) The energy given out can be calculated using the formula:

energy given out = mass of water × 4.2 × temperature change

Use the data given to calculate the energy given out when burning methanol. [2]

Energy given out = ... J

(iii) Give the **letter** of the correct conclusion for the student's investigation. [1]

A all alcohols burn giving out the same amount of energy

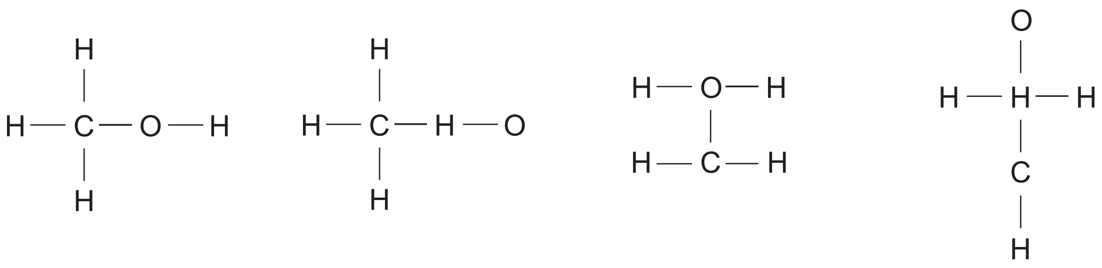
B the greater the number of carbon atoms in the alcohol molecule the less energy is given out

C the greater the number of carbon atoms in the alcohol molecule the more energy is given out

D as the number of carbon atoms doubles the amount of energy given out doubles

Letter ....

(iv) Give the **letter** of the structural formula of methanol, CH_3OH. [1]



A **B** **C** **D**

Letter ....

3410U201 09

10